



A quasi-zero-stiffness vibration isolator inspired by Kresling origami

Haodong Zhou^{a,1}, Jiangjun Gao^{b,1}, Yao Chen^{a,c,*}, Zhengliang Shen^b, Hengzhu Lv^d, Pooya Sareh^{e,f}

^a Key Laboratory of Concrete and Prestressed Concrete Structures of Ministry of Education, and National Prestress Engineering Research Center, Southeast University, Nanjing 211189, China

^b China Construction Fifth Engineering Bureau Co., Ltd., Changsha 410019, China

^c School of Civil Engineering, Southeast University, Wuxi Campus, Wuxi 214082, China

^d Nanjing Kingdom Architecture Design Co., Ltd., Nanjing 210019, China

^e Creative Design Engineering Lab (Cdel), School of Engineering, Newcastle University, Newcastle upon Tyne NE1 7RU, United Kingdom

^f Escuela Técnica Superior de Ingeniería y Diseño Industrial, Universidad Politécnica de Madrid (UPM), Madrid 28012, Spain

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ABSTRACT

Vibration isolators are widely used to mitigate undesirable vibrations in engineering systems and structures, but low-frequency vibration isolation remains a challenge. Origami-inspired structures, with their tunable stiffness and transformative capabilities, offer potential for vibration isolation. Inspired by conical Kresling origami, this study proposes a quasi-zero-stiffness vibration isolation system. Static analysis of the origami structure reveals that the system exhibits near-zero dynamic stiffness at the static equilibrium position, while maintaining static stiffness during the load-bearing process. The amplitude-frequency characteristic formula and vibration transmissibility formula are derived using the harmonic balance method. The effects of varying damping ratios and excitation amplitudes on the amplitude-frequency and transmissibility curves are examined. It is shown that the vibration isolation range increases with the number of layers. A comparison of the system's response to harmonic vibrations under different compressive deformations highlights the effectiveness of the origami quasi-zero-stiffness system as a vibration attenuator. Compared to existing quasi-zero-stiffness dampers, the origami-inspired system features a wider isolation frequency band, a lower initial isolation frequency, and reduced vibration transmissibility, demonstrating superior isolation performance, particularly at low frequencies.

1. Introduction

Vibration isolators have been widely employed to mitigate adverse vibrations in engineering systems [1–5]. However, the low-frequency isolation performance of linear isolators is limited, as excitation frequencies must significantly exceed their natural frequency [6,7]. To address this issue, the quasi-zero-stiffness (QZS) isolator has garnered significant attention for its superior low-frequency isolation performance [8–10].

1.1. Advantages of QZS isolators

The differences between traditional linear isolators and QZS isolators are illustrated in Fig. 1. As shown in Fig. 1(a), the stiffness of a linear

isolator remains constant, whereas the QZS isolator exhibits a unique high-static, low-dynamic stiffness behavior. The dynamic stiffness, defined as the first derivative of the reaction force with respect to displacement, is intentionally designed to be small or even zero near the static equilibrium point. In contrast, static stiffness—referring to the ratio of the reaction force at the static equilibrium position to the displacement from the initial position—can be made very large. Fig. 1(b) shows the effective isolation range for both systems, where the frequency ratio is the ratio of the operating frequency to the natural frequency, and ω_0 and ω'_0 represent the starting frequency ratios for vibration isolation. In the effective isolation region, the frequency ratio for the linear isolator must exceed $\sqrt{2}$. In contrast, the QZS isolator, due to its high-static, low-dynamic stiffness behavior, can reduce the natural frequency and extend the isolation band into the low-frequency range

* Corresponding author at: Key Laboratory of Concrete and Prestressed Concrete Structures of Ministry of Education, and National Prestress Engineering Research Center, Southeast University, Nanjing 211189, China.

E-mail address: chenyao@seu.edu.cn (Y. Chen).

¹ The authors Zhou and Gao contribute equally to this paper.

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while maintaining a high load-bearing capacity.

1.2. Applications of QZS isolators

QZS isolators have been widely applied across various fields, including automotive engineering [11], pipeline engineering [12], and civil engineering [13]. Vibration control in bridges is a critical issue in civil engineering. Bouna et al. [13] studied the vibration control performance of multiple QZS isolators installed under the piers of a multi-span continuous bridge. They demonstrated that QZS isolators can effectively suppress bridge vibrations caused by moving loads [14]. Attary et al. [15] proposed a QZS isolation system to reduce low-frequency vibrations in bridges, with shake table experiments and simulations verifying its excellent isolation performance. QZS systems have also been applied in seismic isolation devices to mitigate earthquake hazards. Eskandary et al. [16] designed a spring-linked QZS seismic model that effectively reduces the transmission of seismic acceleration and forces. Li et al. [17] combined a QZS seismic structure with an active control system to adapt to various types of earthquakes. Cimellaro et al. [18] developed a three-directional QZS isolation system to control both the horizontal and vertical components of seismic motion during strong earthquakes. Zhou et al. [19] integrated a QZS system for vertical isolation with a friction pendulum system for horizontal isolation.

1.3. Components of QZS isolators

A typical QZS system comprises a parallel combination of positive stiffness structures and negative stiffness mechanisms [20,21]. Linear springs are typically used as the positive stiffness component, while negative stiffness structures play a crucial role in shaping the characteristics of QZS systems.

Inclined spring systems [22,23] and buckling beams [24] serve as simple mechanisms for generating negative stiffness. Zhao et al. [25] proposed a novel QZS system incorporating three pairs of inclined springs, with stiffness formulas derived and experimentally validated. Additionally, cam-roller systems provide a compact design for negative stiffness in both axial and torsional directions [26,27], while magnetic spring systems [28–31] offer a non-contact design to minimize friction. Yan et al. [32] achieved QZS characteristics using magnetic springs and analyzed their vibration isolation performance both numerically and experimentally. X-shaped structures [33,34] and biomimetic designs [35–37] ingeniously combine rods and springs to produce systems with impressive vibration isolation performance. Gatti et al. [38] introduced a muscle-inspired isolator exhibiting QZS characteristics at larger deflections, and Chai et al. [39] proposed a 3-degree-of-freedom damping unit based on an X-shaped structure that achieved low-frequency isolation in three directions.

Origami structures, which can take on various configurations

through different folding patterns, exhibit a range of mechanical properties such as multi-stability [40,41], negative Poisson's ratio [42,43], reconfigurability [44,45], and adjustable stiffness [46]. Leveraging these remarkable characteristics, origami structures have recently been applied in sound and vibration control [47,48]. Zhang et al. [49] experimentally validated the effective bandgap of perforated Miura-ori phononic structures, while Ji et al. [50] reviewed recent developments in the application of origami structures for vibration control. They also introduced a stacked Miura-ori structure providing nonlinear stiffness and discussed the role of damping [51]. Liu et al. [52] investigated the negative stiffness properties of Tachi-Miura origami and applied it to QZS isolation systems. However, due to the complex mathematical models and strong nonlinearity associated with certain configurations, the application of origami structures in vibration isolators is still in its early stages.

Inspired by the conical Kresling origami, we propose a novel passive isolator consisting of two symmetric conical Kresling origami units and an axial spring. The negative stiffness is tuned by adjusting the geometric parameters of the origami structure, while the axial spring with positive stiffness serves as compensation to maintain system stability, thereby achieving the QZS characteristics. The static performance of the QZS system is verified by analyzing force-displacement responses and stiffness characteristics. Displacement transmissibility is then derived through dynamic analysis using the Harmonic Balance Method and numerical simulations. The impact of system damping on isolation performance is also discussed. The isolation performance under various parameters will be evaluated to confirm the advantages of the proposed isolator.

2. Design and static behavior of an origami-inspired QZS isolator

2.1. Detailed design of the vibration isolator

The proposed conical Kresling origami vibration isolator (CKOVI) is illustrated in Fig. 2, with its geometric shape detailed in Fig. 2(a). The structure consists of two symmetric conical Kresling origami units and an axial spring. The conical Kresling origami unit, which provides negative stiffness, is shown alongside its folding pattern and geometric configuration in Fig. 2(b).

Notably, the top and bottom planes of the conical Kresling origami structure can rotate relative to each other without affecting the radii of the top and bottom planes or the structural height. The geometric configuration of the conical Kresling origami structure is determined by five independent initial parameters: the top plane radius R_1 , the bottom plane radius R_2 , structural height h , relative rotation angle θ between the upper and lower bases, and the number of edges n of the conical Kresling origami structure. Due to the rotational symmetry of the structure, all lengths of the mountain and valley creases, as well as their

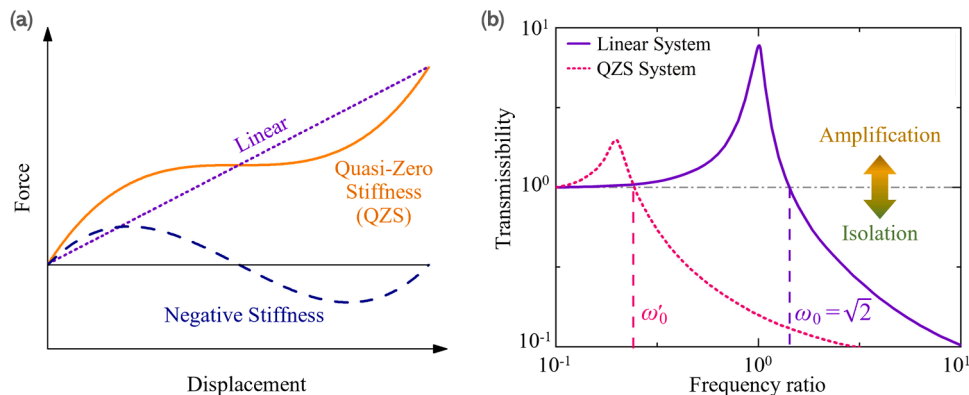


Fig. 1. Comparison between linear isolator and QZS isolator: (a) Force-displacement curve. (b) Transmissibility response.

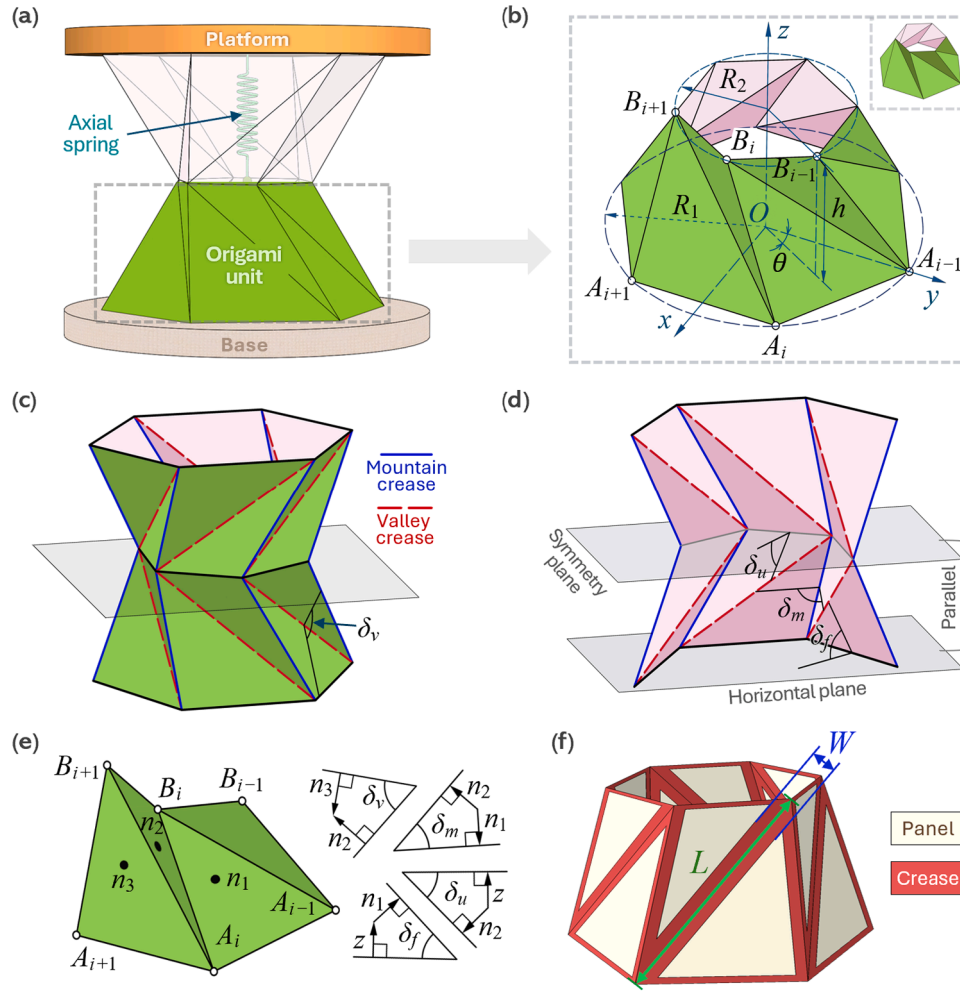


Fig. 2. Schematic of the proposed origami isolator: (a) Composition of the vibration isolator. (b) A typical conical Kresling origami unit. (c) Symmetrical conical Kresling origami structure. (d) Section view of the conical Kresling origami structure. (e) Folding Angles. (f) Crease design.

respective folding angles, are uniform. The conical Kresling origami structure functions as a dual-degree-of-freedom system, with h and θ chosen as independent variables. A Cartesian coordinate system is established, as shown in Fig. 2(b), where O is the origin, A_i denotes the vertex of the bottom polygon, and B_i denotes the vertex of the top polygon. During the folding process, the coordinates of each vertex can be expressed as

$$A_i = \begin{pmatrix} R_1 \cos\left(\frac{2\pi i}{n}\right) & R_1 \sin\left(\frac{2\pi i}{n}\right) & 0 \end{pmatrix}, \forall i \in \{0 \dots n-1\} \quad (1)$$

$$B_i = \begin{pmatrix} R_2 \cos\left(\frac{2\pi i}{n} + \theta\right) & R_2 \sin\left(\frac{2\pi i}{n} + \theta\right) & h \end{pmatrix}, \forall i \in \{0 \dots n-1\} \quad (2)$$

Vectors $\vec{OA}_i$ and $\vec{OB}_i$ represent the position vectors of vertices A_i and B_i , respectively. Vectors $\vec{l}_{A_i B_i}$ and $\vec{l}_{A_i B_{i+1}}$ denote the vectors of mountain and valley creases, respectively. They should satisfy

$$\vec{l}_{A_i B_i} = \vec{OB}_i - \vec{OA}_i, \quad \text{and} \quad \vec{l}_{A_i B_{i+1}} = \vec{OB}_{i+1} - \vec{OA}_i \quad (3)$$

To calculate the dihedral angles formed by the creases, it is necessary to determine the normal vectors of the triangular panels $A_i B_i A_{i-1}$, $A_i B_{i+1} B_i$, and $A_i A_{i+1} B_{i+1}$. The normal vector of panel $A_i B_i A_{i-1}$ is $\vec{n}_1$, the normal vector of panel $A_i B_{i+1} B_i$ is $\vec{n}_2$, and the normal vector of panel $A_i A_{i+1} B_{i+1}$ is $\vec{n}_3$, given by

$$\vec{n}_1 = \vec{l}_{A_i A_{i-1}} \times \vec{l}_{A_i B_i}, \quad \vec{n}_2 = \vec{l}_{A_i B_i} \times \vec{l}_{A_i B_{i+1}}, \quad \vec{n}_3 = \vec{l}_{A_i B_{i+1}} \times \vec{l}_{A_i A_{i+1}} \quad (4)$$

The dihedral angles for the mountain crease, valley crease, upper base crease, and lower base crease are denoted by δ_m , δ_v , δ_u , and δ_f , respectively. The schematic of the folding angles is shown in Fig. 2(c)-(d). The expressions for these dihedral angles can be derived from the normal vectors $\vec{n}_1$, $\vec{n}_2$, and $\vec{n}_3$:

$$\begin{aligned} \delta_m &= \pi - \angle(\vec{n}_1, \vec{n}_2), \quad \delta_v = \pi - \angle(\vec{n}_2, \vec{n}_3), \quad \delta_f = \angle(\vec{n}_1, \vec{z}), \quad \delta_u \\ &= \pi - \angle(\vec{n}_2, \vec{z}) \end{aligned} \quad (5)$$

The origami structure consists of two types of components: creases and panels. Flexible creases allow the structure to exhibit additional degrees of freedom and enable complex deformations such as stretching and twisting. Using flexible materials for the creases can reduce stress on these components and improve the fatigue life of the structure [53]. To achieve these complex deformations, the crease design is performed with a width-to-length ratio (i.e., W/L) of 0.1 for each crease [54], as shown in Fig. 2(f). Appropriate dimensions enable the origami structure to effectively guide the deformation path under applied loads and achieve the desired mechanical performance [55]. The initial parameters are set as $R_1 = 20$ mm, $R_2 = 30$ mm, $h_0 = 38.2$ mm, and a twist angle $\theta_0 = 1.22$ rad. Note that these parameters represent one set of values among many possible options; other values can also achieve the bistable characteristics of the designed structure [55,56]. In the simulation

process, the selection of panel materials balances stiffness and mass, ensuring that the structure is lightweight yet sufficiently stiff [57]. For the creases, material selection focuses on flexibility and durability to ensure effective elastic recovery and realistic simulation of the folding process [58,59]. Therefore, the material of the panels is nylon with the following properties: $E = 1.1 \times 10^3$ MPa, $\mu = 0.42$, and $\rho = 1.13$ g/cm³, where E is the modulus of elasticity, μ is the Poisson's ratio, and ρ is the density of the material. The crease material is the flexible thermoplastic polyurethane material with $E = 12$ MPa, $\mu = 0.42$, and $\rho = 1.12$ g/cm³ [57]. The isolator is composed of thin facets and planes made of lightweight material and can be considered massless relative to the supported load.

2.2. Multi-stability and negative stiffness of the origami unit

Previous studies [60,61] have considered two types of origami configurations: the full model, which includes all creases and panels [62], and the equivalent truss structure [63]. Within the equivalent truss model, there are two subtypes: the bar-hinge model [53,64] and the truss model [65,66]. The bar-hinge model represents the stretching and shear of the origami panel with elastic bars and the folding of creases with elastic hinges, while the truss model further simplifies by neglecting the folding of creases [67,68,70,71]. Given that the thickness of the origami panel is typically much smaller than its in-plane dimensions, finite element models based on shell elements are used [69]. Fig. 3 illustrates three common mechanical models of origami-inspired structures. A comparison among the truss model, bar-hinge model, and thin-shell model has been conducted using finite element simulations to analyze the quasi-static mechanical behavior of the structure.

Based on the deformation characteristics of non-rigid origami structures, the total potential energy during the folding process is composed of the strain energy generated by the deformations of creases and panels. The elastic potential energy at the creases, U_c , represents the strain energy stored in the creases. The initial angle of the creases is δ_i^0 , which changes to δ_i during folding. The strain and stress in the creases can be derived based on the principles of elastic mechanics. Therefore, the elastic potential energy, U_c , induced by nonlinear folding is given by

$$U_c = n \sum_{i=v,m,b,f} \left(\frac{1}{2} k_c (\delta_i - \delta_i^0)^2 \right), \quad k_c = \frac{E_c I_c}{W_c}, \quad I_c = \frac{L_c t_c^3}{12} \quad (6)$$

where E_c is the Young's modulus of the crease material, L_c is the length of the crease, t_c is the thickness of the crease, and W_c is the width of the crease.

The nonlinear geometry and kinematics of the Kresling origami structure make it challenging to calculate panel deformations. To simplify this, the deformations can be approximated as stretching deformations of the creases, treating them as linear springs. Consequently, the elastic potential energy stored in the panels can be equated to the

elastic potential energy in linear springs. Since the upper and lower bases are rigid, only the mountain and valley creases experience stretching deformations. The initial lengths of the mountain and valley creases in a conical Kresling structure are denoted as a_0 and b_0 , respectively. Thus, the potential energy U_p generated by the panel deformations in the conical Kresling origami structure can be expressed as

$$U_p = \frac{1}{2} n k_m \left(\left\| \vec{l}_{A_i B_i} \right\| - a_0 \right)^2 + \frac{1}{2} n k_v \left(\left\| \vec{l}_{A_i B_{i+1}} \right\| - b_0 \right)^2 \quad (7)$$

where k_m and k_v represent the stiffness of the equivalent springs corresponding to the mountain and valley creases, respectively. The total elastic potential energy U can be obtained by

$$U = U_p + U_c \quad (8)$$

Based on the principle of energy and Eqs. (1) to (8), the total potential energy of the conical Kresling structure during folding can be evaluated. Suitable initial geometric parameters can enable the structure to exhibit multi-stable behavior. For the QZS isolator, a bistable configuration is selected to provide the necessary negative stiffness.

Fig. 4(a) shows the quasi-static mechanical behavior of the truss model, bar-hinge model, and thin-shell model as analyzed using finite element methods. Discrepancies between the results may stem from the truss model's assumption of no contact between panels during unfolding, whereas panel contact is accounted for in the finite element simulations. Furthermore, the use of flexible materials for the creases, combined with more rigid panel materials, can cause torsional deformations at the creases—a behavior that the truss model cannot simulate. Therefore, we will focus on the finite element method (FEM) results based on the thin-shell model for further investigation.

When the thickness of the structure ranges from 0.3 mm to 0.6 mm, the corresponding force-displacement curves are obtained and shown in Fig. 4(b). It can be observed that the thickness has almost no effect on the region where the structure exhibits negative stiffness. A thickness of 0.5 mm is therefore chosen for further investigation. Fig. 5(a) illustrates the deformation of the conical Kresling origami unit during the folding process, with the axial compressive displacement of the structure denoted by Δ . It can be observed that the strain at the crease increases significantly during folding. Fig. 5(b) depicts the Von-Mises stress distribution in the conical Kresling origami unit. The results show that the creases do not carry a substantial amount of stress, indicating that the panels are primarily responsible for bearing the Von-Mises stress. These dual phenomena work together to facilitate the folding process effectively.

2.3. QZS mechanism of the origami-inspired vibration isolator

Although the energy landscape of the conical Kresling unit is highly nonlinear and complex, the conical Kresling structure exhibits bistability. During the deformation process, the structure undergoes both

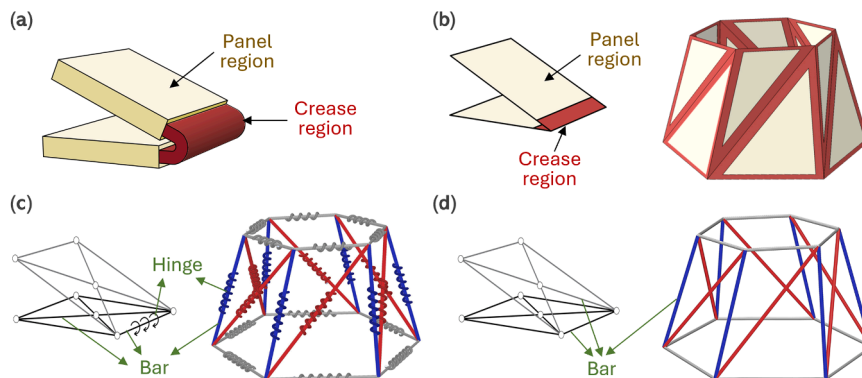


Fig. 3. Mechanical models of origami-inspired structures: (a) Real origami-inspired structure. (b) Thin-shell model. (c) Bar-hinge model. (d) Axial truss model.

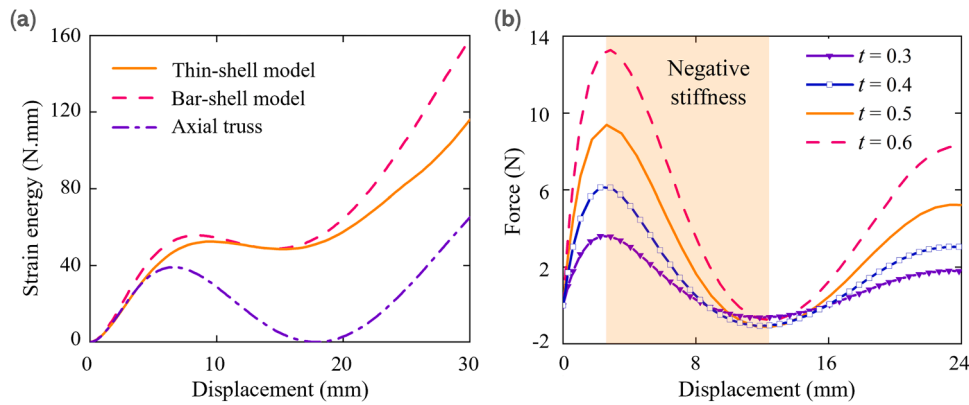


Fig. 4. Static behavior of the conical Kresling origami model: (a) Strain energy behavior for different numerical models. (b) Effect of thickness t on static characteristics.

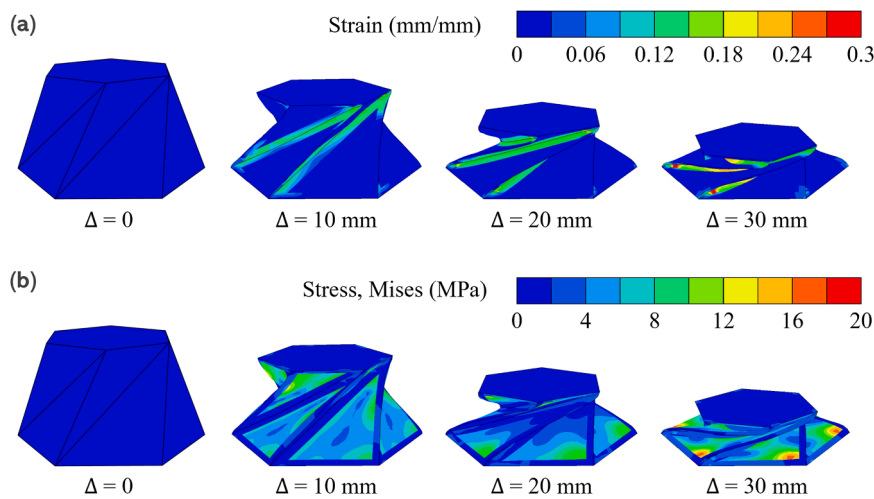


Fig. 5. Folding process of a conical Kresling unit with $t = 0.5$ mm: (a) Strain distribution. (b) Von-Mises stress distribution.

axial deformation and rotation. Therefore, the expression for the potential energy U can be given by

$$U = u(h, \theta) \quad (9)$$

The nonlinear reaction force F and reaction moment T for the conical Kresling structure are derived by taking the first-order derivatives of the total potential energy function, expressed as

$$F = \frac{dU}{dh}, \quad T = \frac{dU}{d\theta} \quad (10)$$

For the base with free rotation, the expression of the function $\theta = f(h)$ can be determined. The conical Kresling structure can be simplified as a single-degree-of-freedom system with the variable h . The axial stiffness of the structure, K , can be expressed as

$$K = \frac{dF}{dh} \quad (11)$$

Through the above analysis, the introduction of an axial spring ($k_s = 1.28$ N/mm) adjusts the positive stiffness to match the minimum negative stiffness, as shown in Fig. 2(a). This adjustment results in a vibration isolator with quasi-zero-stiffness (QZS) properties based on the conical Kresling structure. Additionally, different gravitational loads can be accommodated by adjusting the precompression of the axial spring. The potential energy of the designed origami vibration isolator, which consists of the base and the axial spring with positive stiffness, is given by

$$U_e = U + \frac{1}{2}k_s(h - L_s)^2 \quad (12)$$

where k_s represents the stiffness of the spring, and L_s denotes the initial length of the spring. Similar to Eqs. (10) and (11), the reaction forces and axial stiffness of the origami structure can be derived using Eq. (12). The expressions are

$$F_e = \frac{dU_e}{dh} = \frac{dU}{dh} + k_s(h - L_s), \quad K_e = \frac{dF_e}{dh} = \frac{dF}{dh} + k_s \quad (13)$$

According to the simulation results in Section 2.2, the axial spring was integrated into the conical Kresling unit, and quasi-static compression tests were conducted. Fig. 6 illustrates the relationship between reaction force and displacement for a basic cell of the structure, as well as the relationship between stiffness and displacement. The stiffness shown in Fig. 6 is calculated as the first derivative of the reaction force with respect to displacement, representing the dynamic stiffness. It can be observed that the dynamic stiffness of the structure is sufficiently low, activating the QZS characteristic only under specific loading conditions (the shaded area in Fig. 6). Besides, the stiffness response is asymmetric around the minimum stiffness position, which occurs at a compression displacement of 6 mm. The segment near the designed static equilibrium position features zero stiffness, while the dynamic stiffness remains quite low. This configuration makes it suitable for vibration isolation under dynamic loads.

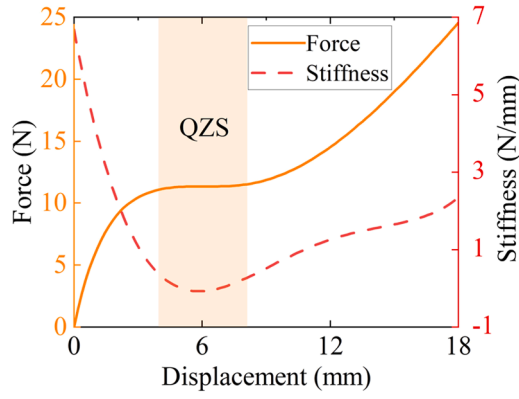


Fig. 6. Force-displacement and stiffness-displacement curves of the Kresling origami unit.

3. Dynamic modeling and analysis

3.1. Equation of motion

In dynamic analysis, due to the massless nature of the neutral plane of the conical Kresling structure, the relationship between θ and h remains unchanged. In other words, Eq. (13) for the vertical reaction force remains applicable. Since vibrations predominantly affect the vicinity of the equilibrium point, dynamic system control equations formulated under static equilibrium conditions are employed. Additionally, only small deflections of the basic unit are considered. The expression for the vertical reaction force of the isolator can be approximated using a polynomial series:

$$\bar{F} = K_1 \bar{h} + K_2 \bar{h}^3 + K_3 \bar{h}^5 + \dots + K_p \bar{h}^p \quad (14)$$

where $\bar{h} = h - h_0$ represents the displacement after coordinate transformation, h_0 is the height of the isolator at the stable equilibrium, $\bar{F} = F - F_0$ is the force after coordinate transformation, F_0 is the force of the isolator at the stable equilibrium, and p represents the highest order of the fitting polynomial. It is noted that a higher order p lead to more accurate fitting results. Thus, for satisfactory accuracy, $p = 5$ is adopted in this study. The fitting result is $K_1 = 0$, $K_3 = 0.02478 \text{ N/mm}^3$, and $K_5 = 7.3619 \times 10^{-5} \text{ N/mm}^5$. As shown in Fig. 7, the fitting results align well with the original data near the static equilibrium, and the fitting accuracy meets the requirements. As the displacement amplitude increases, the error between the two gradually grows. However, the overlap within one-third of the total effective displacement range is acceptable and satisfies practical engineering needs.

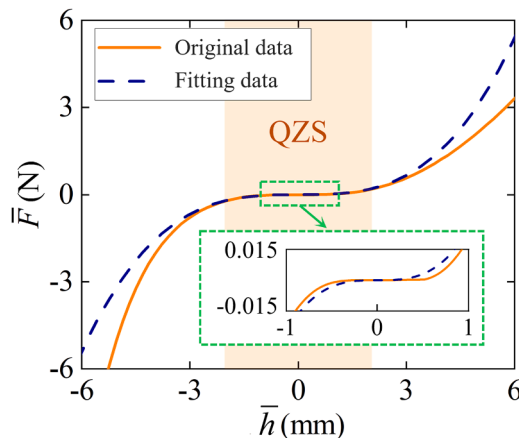


Fig. 7. Fitting result for the vertical reaction force.

Assuming that the isolation system is subjected to an excitation at y , and the absolute displacement of the isolated object is denoted as z , the relevant displacement can be expressed as $u = z - y$, as shown in Fig. 8. The origami isolator can be viewed as a combination of equivalent springs and equivalent damping. With the mass of the load as m and choosing its absolute displacement as the generalized coordinate, we can describe the kinetic energy E_k of the entire system as

$$E_k = \frac{1}{2} m \dot{z}^2 \quad (15)$$

The potential energy V_m of the origami isolator system must take into account the potential energy of the isolated object,

$$V_m = U_e + mgz = U + \frac{1}{2} k_s (h - L_s)^2 + mgz \quad (16)$$

Considering the damping effect of the isolation system, Rayleigh's dissipation function can be written as

$$\bar{\Psi} = \frac{1}{2} c \dot{u}^2 = \frac{1}{2} c (\dot{z} - \dot{y})^2 \quad (17)$$

where c represents the linear viscous damping coefficient of the structure and the dissipation function $\bar{\Psi}$ indicates the dissipation rate of the mechanical energy of the damped system. Through Lagrange's equation, the control equation for the isolation system is obtained, given by

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{z}} \right) - \frac{\partial T}{\partial z} + \frac{\partial V}{\partial z} + \frac{\partial \Psi}{\partial \dot{z}} = 0 \quad (18)$$

Through Eqs. (15) to (18), the motion equation can be expressed as

$$m \ddot{z} + F + mg + c(\dot{z} - \dot{y}) = 0 \quad (19)$$

This dynamic model is established based on the initial position. For the isolation system, when the vertical reaction force of the isolator is directly balanced by the gravity of the isolated object (i.e., $F_0 + mg = 0$), the expected dynamic response under static equilibrium can be rewritten as

$$m \ddot{z} + F - F_0 + c(\dot{z} - \dot{y}) = 0 \quad (20)$$

As the simplified force model has been described, the dynamic model of the structure can be obtained by substituting Eq. (14) into Eq. (20),

$$m \ddot{u} + c \dot{u} + K_3 \bar{u}^3 + K_5 \bar{u}^5 = -m \ddot{y} \quad (21)$$

where $\bar{u} = u - u_0$ is the relevant displacement after the coordinate transformation, and u_0 is the displacement of the isolator at the stable equilibrium.

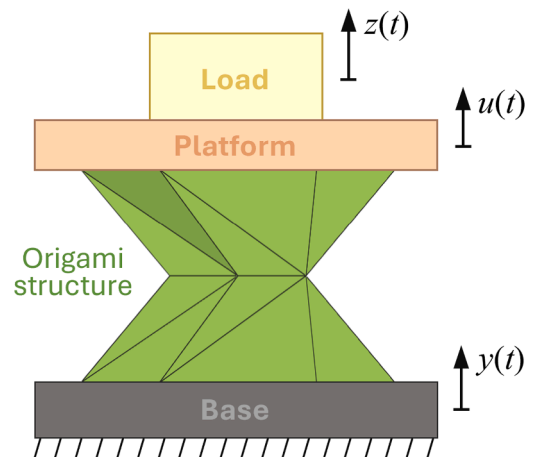


Fig. 8. Dynamic model of the origami-inspired vibration isolation system in the static equilibrium state.

3.2. Steady-state response

When the system is subjected to harmonic displacement excitation (i.e., $y = Y \cos(\omega t)$), the following dimensionless parameters are introduced, given by

$$\omega_0 = \sqrt{\frac{k_s}{m}}, \quad \xi = \frac{c}{2m\omega_0}, \quad \Omega = \frac{\omega}{\omega_0}, \quad \tau = \omega_0 t, \quad \hat{Y} = \frac{Y}{R_1},$$

$$\hat{x} = \frac{\bar{u}}{R_1}, \quad \dot{\hat{x}} = \frac{\dot{\bar{u}}}{R_1\omega_0}, \quad \ddot{\hat{x}} = \frac{\ddot{\bar{u}}}{R_1\omega_0^2}, \quad \gamma_3 = \frac{K_3}{k_s}R_1^2, \quad \gamma_5 = \frac{K_5}{k_s}R_1^4 \quad (22)$$

The motion equation can be expressed as

$$\ddot{\hat{x}} + 2\xi\dot{\hat{x}} + \gamma_3\hat{x}^3 + \gamma_5\hat{x}^5 = \hat{Y}\Omega^2 \cos(\Omega\tau) \quad (23)$$

The harmonic balance method is employed to determine the primary resonance response. Considering only harmonic excitation, the solution to Eq. (21) can be expressed as

$$\hat{x} = A \cos(\Omega\tau + \varphi) = A \cos \varphi \quad (24)$$

where Ω is the dimensionless frequency and A is the amplitude of the structural response. Therefore, Eq. (23) can be rewritten as

$$-\Omega^2 A \cos \varphi - 2\xi A \Omega \sin \varphi + \gamma_3 A^3 \cos^3 \varphi + \gamma_5 A^5 \cos^5 \varphi = \hat{Y}\Omega^2 \cos(\varphi + \phi) \quad (25)$$

By neglecting higher harmonic terms using trigonometric functions and angle formulas, and setting the coefficient of the first-order term to zero, the expression can be simplified as

$$\begin{cases} \hat{Y}\Omega^2 \cos \varphi = -A\Omega^2 + \frac{3}{4}\gamma_3 A^3 + \frac{5}{8}\gamma_5 A^5 \\ \hat{Y}\Omega^2 \sin \varphi = 2\xi A \Omega \end{cases} \quad (26)$$

By summing the squares and eliminating phase angles, the amplitude-frequency relationship under the base excitation can be determined by

$$\left(-\Omega^2 A + \frac{3}{4}\gamma_3 A^3 + \frac{5}{8}\gamma_5 A^5 \right)^2 + (2\xi A \Omega)^2 = (\hat{Y}\Omega^2)^2 \quad (27)$$

Similarly, when a harmonic excitation force $f = F_t \cos(\omega t)$ is applied to the isolator, the forced vibration equation for the system is

$$\ddot{\hat{x}} + 2\xi\dot{\hat{x}} + \gamma_3\hat{x}^3 + \gamma_5\hat{x}^5 = \hat{F} \cos(\Omega\tau) \quad (28)$$

where F_t represents the amplitude of the input harmonic excitation force, and $\hat{F} = \frac{F}{R_1\omega_0^2 m}$ denotes the dimensionless force. From Eq. (28) and the derivation process described above, the amplitude-frequency curve can be obtained by

$$\left(-\Omega^2 A + \frac{3}{4}\gamma_3 A^3 + \frac{5}{8}\gamma_5 A^5 \right)^2 + (2\xi A \Omega)^2 = \hat{F}^2 \quad (29)$$

The structural amplitude-frequency characteristic curve under harmonic excitation force is shown in Fig. 9. As the damping ratio ξ increases, the curvature of the amplitude-frequency characteristic curve gradually decreases. This trend makes the QZS isolation system's response more similar to that of a linear system.

4. Results and discussions

4.1. Vibration isolation performance

The absolute displacement transmissibility can measure the performance of the proposed isolator. The transmissibility of absolute displacement T_d^a under the excitation of the isolator is defined as the ratio of the absolute displacement amplitude $\text{Am}[\hat{z}]$ on the foundation to the absolute displacement amplitude $\text{Am}[\hat{y}]$ on the isolator input. In general, a smaller absolute displacement transmissibility corresponds to a more effective capabilities of vibration suppression. For comparison, T_d^a can be expressed in decibel (dB) form

$$T_d^a = 20 \lg \frac{\text{Am}[\hat{z}]}{\text{Am}[\hat{y}]} = 20 \lg \frac{\sqrt{A^2 + \hat{Y}^2 + 2A\hat{Y} \cos \varphi}}{\hat{Y}} \quad (30)$$

where φ is the initial phase angle and can be solved according to Eq. (26).

To visually demonstrate the isolation performance of the proposed QZS isolator, harmonic displacement loads are applied to the top plate based on the finite element model established in Section 2.3. The time history curves of the reaction forces on the top plate and the bottom plate are presented in Fig. 10(a). Additionally, random vibration loads are applied to the top plate, with the time-history curves of the reaction forces on both plates shown in Fig. 10(b). The significant reduction in the output force amplitudes compared to the input counterparts highlights the effectiveness of the proposed QZS structure in isolating vibrations.

4.2. Comparison with existing QZS isolators

The most widely used QZS isolator is perhaps the Three-Spring Vibration Isolator (TSVI), which has been extensively studied [13,14]. Therefore, we have selected the TSVI as a reference for comparison. To ensure a fair evaluation of isolation performance between the CKOVI and TSVI, we have used isolators with the same load mass and natural frequency. The force-displacement curve for a traditional three-spring QZS system is given by

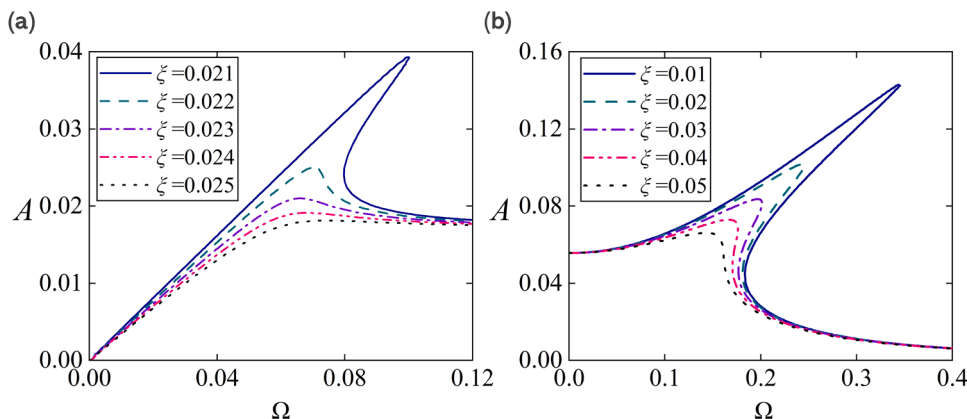


Fig. 9. Amplitude-frequency curves for different damping ratios: (a) System response under displacement excitation. (b) System response under force excitation.

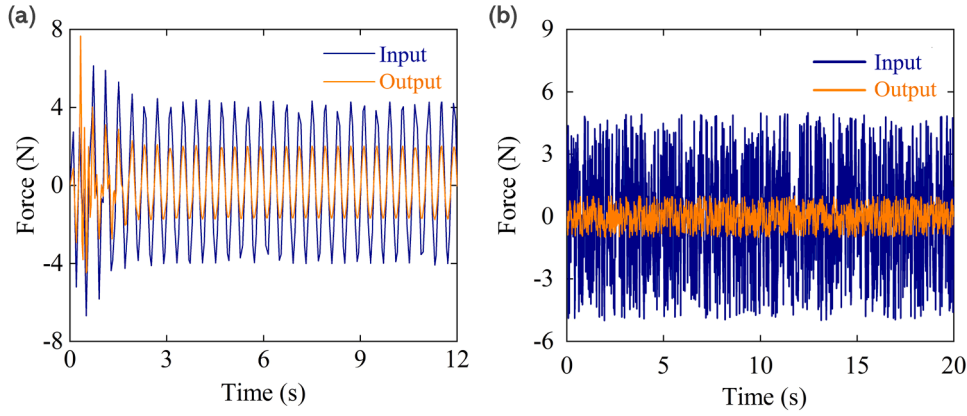


Fig. 10. Dynamic behavior of the QZS isolator: (a) Time history curves under harmonic excitation. (b) Time history curves under random vibration excitation.

$$f = k_v x + 2k_h \left(1 - \frac{L_{in}}{\sqrt{x^2 + L^2}} \right) x \quad (31)$$

It can be rewritten as

$$f = \left[k_v - 2 \left(\frac{L_{in}}{L} - 1 \right) k_h \right] x + \frac{L_{in}}{L^3} K_h x^3 = K_1 x + K_3 x^3 \quad (32)$$

The damped forced vibration equation can be expressed as

$$m\ddot{x} + c\dot{x} + K_1 x + K_3 x^3 = -m\ddot{y}_h = m\omega^2 X_h \cos \omega t \quad (33)$$

The solution process for Eq. (33) follows the same derivation procedure outlined in Section 3.2.

With identical mass, natural frequency, and damping coefficients for both the CKOVI and the TSVI, the transfer efficiency for these two isolators is depicted in Fig. 11. Both isolators exhibit significant isolation effects. Due to their shared QZS characteristics, their isolation performance remains nearly identical at high frequencies. However, the Kresling isolator demonstrates a substantially lower effective initial isolation frequency compared to the traditional system, resulting in a broader isolation bandwidth. Consequently, the CKOVI shows significant potential as a high-performance low-frequency isolator in civil engineering applications, such as vibration control for bridges and seismic isolation.

4.3. Programmable characteristics

The vertical stacking layer number N is an important and intuitive structural parameter for folded-type vibration isolators, and these parameters directly determine the range of motion for the proposed dynamic system. The impact of N on displacement transmissibility is

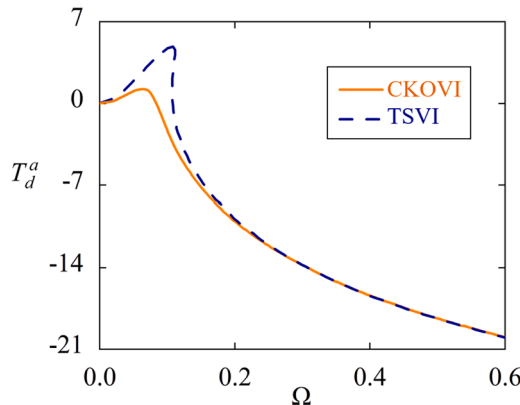


Fig. 11. Comparison of vibration isolation performance between CKOVI and TSVI.

shown in Fig. 12. Displacement transmissibility T is defined as the ratio of the amplitude of the input to the output $T = \text{Am}[\hat{z}]/\text{Am}[\hat{y}]$. With an increase in the number of layers or a decrease in the equivalent stiffness ratio, superior isolation performance can be achieved. Specifically, the effective initial isolation frequency on the horizontal axis shifts to the left, expanding the effective isolation frequency band. The transmissibility curve shifts downward smoothly, without bending, and shows smaller absolute displacements. Additionally, the transmissibility curve declines steadily, with the resonance peak disappearing, indicating that the isolation system can provide beneficial effects across the entire frequency spectrum.

The effects of N on the isolation performance can be explained by the effective QZS range. The relationship between force and displacement for a two-layer stacked QZS system is shown in Fig. 13(a). Compared to a single-layer structure, the plateau force has not increased, but the corresponding effective QZS range has doubled. Fig. 13(b) illustrates the effective QZS range for systems with additional layers. These results are consistent with the fact that CKOVI can be effectively modeled in the vertical direction as a series of N -folded basic modules arranged in parallel.

The effect of damping ratio ξ and excitation amplitude $\hat{Y}$ on displacement transmissibility is shown in Fig. 14. It is evident that increasing the damping ratio or decreasing the excitation amplitude effectively enhances damping performance in the low-frequency range. When the damping ratio increases to 0.04, the displacement transmissibility curve becomes smoother, and response peaks are eliminated. Conversely, when the damping ratio is reduced to 0.02, jumping phenomena and even unbounded responses may occur during the dynamic response process. However, at higher frequencies, the transmissibility curve for higher damping ratios is elevated compared to lower damping

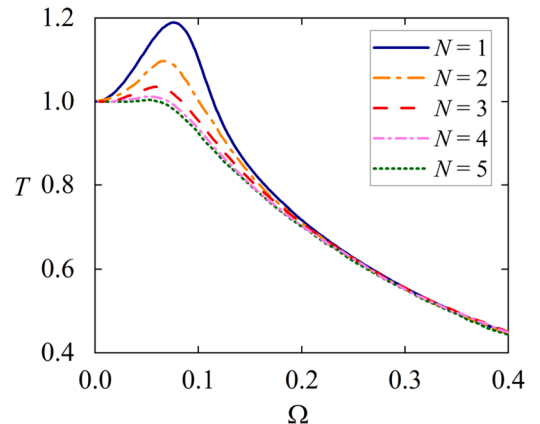


Fig. 12. Transmissibility-frequency curves for different values of N .

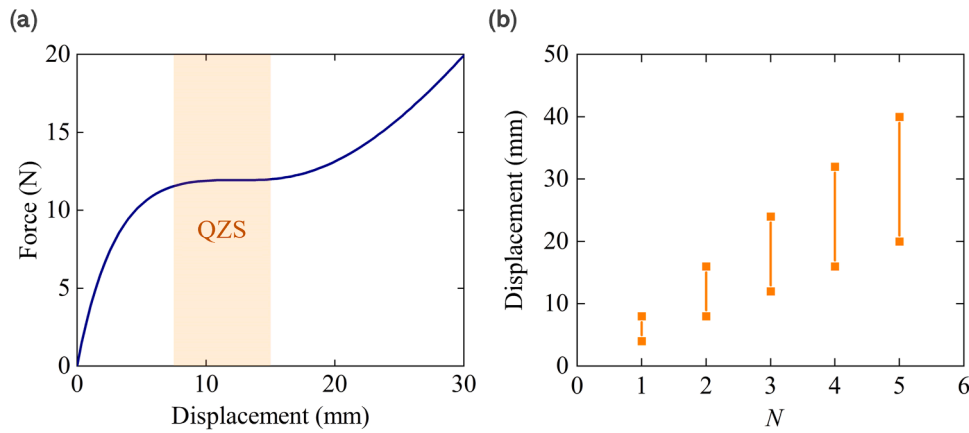


Fig. 13. Effect of N on static characteristics in the vertical direction: (a) Force-displacement curve for a two-story origami structure. (b) QZS locations for different values of N .

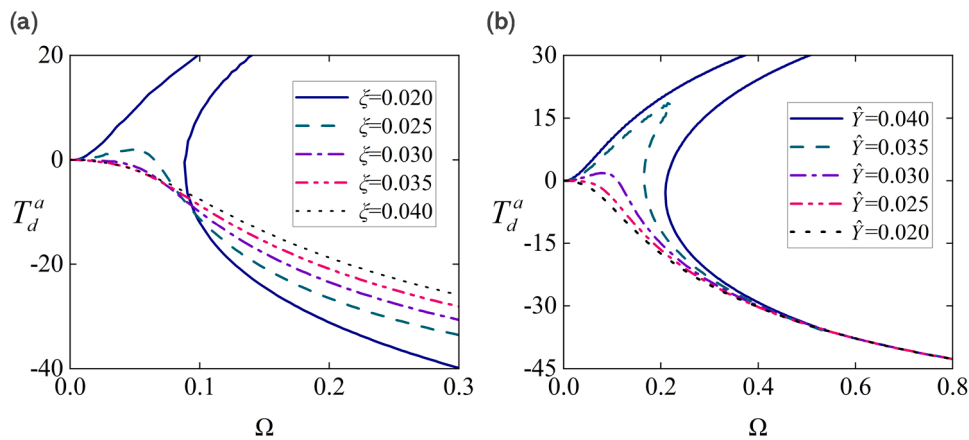


Fig. 14. Relationship between transmissibility and excitation frequency: (a) Effect of damping ratio. (b) Effect of excitation amplitude.

ratios. In other words, higher damping ratios reduce the effectiveness of high-frequency isolation. The influence of damping ratio on this isolator follows a pattern similar to that observed in other nonlinear systems.

The influence of excitation amplitude $\hat{Y}$ on vibration isolation performance shows a relatively straightforward trend. In nonlinear dynamical systems, vibration isolation performance is directly impacted by excitation amplitude. A decrease in excitation amplitude significantly enhances vibration isolation performance and effectively broadens the bandwidth for isolation. When the excitation amplitude surpasses a certain threshold, the dynamic system may experience a jump phenomenon. Moreover, the transmissibility curves tend to converge to a consistent stable value as the excitation amplitude varies.

The transmission efficiency of the structure under different initial compressive deformations (denoted by δ) is shown in Fig. 15. It can be observed that when the initial compressive deformation is $\delta = 6$ mm, the structure is in a QZS state, minimizing transmission efficiency and achieving optimal isolation performance. In other cases, due to the loss of QZS characteristics, the isolation performance is less effective.

5. Conclusions

Based on the conical Kresling origami structure, a nonlinear quasi-zero-stiffness (QZS) vibration isolator with positive stiffness spring compensation has been proposed. The mechanical framework of the proposed origami structure is established according to the geometric configuration of the conical Kresling. An investigation into the principles of potential energy and bistable mechanics is conducted, alongside

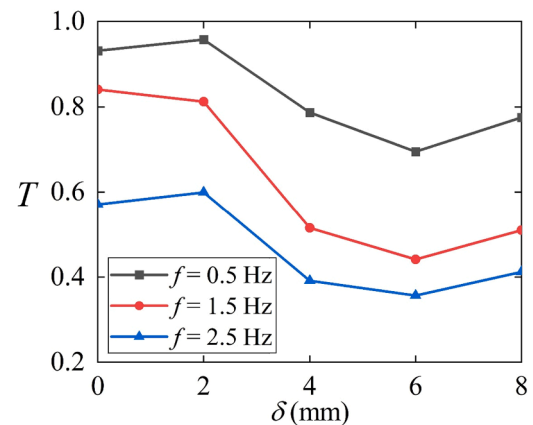


Fig. 15. Transmission efficiency of the structure under different compressive deformations δ .

an analysis of the negative stiffness. The resultant device demonstrates exceptional low-frequency vibration isolation capabilities while maintaining folding performance. Through theoretical modeling, parameter analysis, and comparative assessments against traditional vibration isolators, the efficacy of the proposed origami device in low-frequency vibration isolation is substantiated.

Moreover, the detailed parameter discussion indicates the robust design flexibility and programmability inherent in the origami-inspired

vibration isolator. The isolator can provide comprehensive vibration attenuation across the entire frequency range under conditions of low-intensity base excitation and high damping. Remarkably, the vibration isolation capability can be significantly improved through a series arrangement. Additionally, the performance under random vibrations is assessed. When compared to existing quasi-zero-stiffness damper systems, the origami-based QZS system offers a broader vibration isolation bandwidth, a lower starting isolation frequency, and reduced transmissibility, particularly excelling in low-frequency vibration isolation.

In conclusion, this study presents a feasible approach for low-frequency vibration isolation systems and extends the application of origami-inspired systems. However, similar to most nonlinear systems, the proposed QZS isolator is generally sensitive to parameter uncertainties. Future research will focus on uncertainty quantification and the fabrication of isolation devices to leverage the QZS characteristics and isolation performance.

CRedit authorship contribution statement

Haodong Zhou: Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Yao Chen:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Zhengliang Shen:** Validation, Resources. **Jiangjun Gao:** Validation, Resources. **Pooya Sareh:** Writing – review & editing, Visualization, Funding acquisition, Supervision. **Hengzhu Lv:** Validation, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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